

at the nanoscale. As a result, energy carried by plasmons allows for light localisation in ultra-small volumes – far beyond the diffraction limit of light.

To generate surface plasmons, it is necessary to excite the metal-dielectric interface in which the dielectric constant of the metal is a function of frequency and negative. At the nanoscale, the electromagnetic (EM) field of the EM wave displays the electron cloud due to its well coupling, which is not possible in the case of bulk matter. Hence plasmonics is frequently associated with nanotechnology.

Investigators have found that by creatively designing the metal-dielectric interface, they can generate surface plasmons with the same frequency as the electromagnetic wave but with much smaller wavelength. This phenomenon could allow plasmons to travel along nanoscale wires called ‘interconnects’ in order to carry information from one part of the microprocessor to another.

Methods

Plasmonic waveguides are attracting much attention owing to their ability to operate in various parts of the spectrum – ranging from visible to far-infrared region. A plasmon could travel as far as several micrometres in the slot waveguide (dielectric core with metallic cladding) – far enough to convey a signal from one part of a chip to another. The plasmon slot waveguide squeezes the optical signal, shrinking its wavelength.

Metallic nanowires can provide lateral confinement of the mode below the diffraction limit. Nanowires have larger attenuation than planar films but light transport over a distance of several microns has been demonstrated.

A chain of differently-shaped nanoparticles (such as spheres and rods) can be used to transport EM waves from one nanoparticle to another via the near-field electrodynamic interaction between them. If the second particle is situated in the near field of the

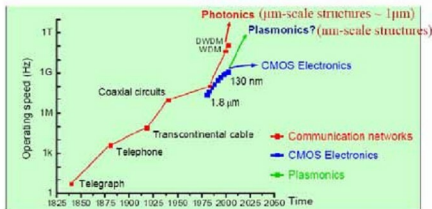


Fig. 1: Operating speed of data transporting and processing systems

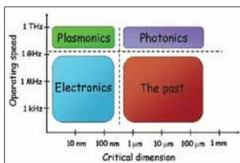


Fig. 2: Operating regimes—applicable size and speed scale—for plasmonic and other devices

other and so on along the chain, EM energy can be propagated within the lateral size confinement less than the diffraction limit. In a chain of closely-spaced nanostructures, the propagation distance depends upon the shape and nature of materials, separation between them as well as the dielectric constant of the host medium.

Latest developments

The possibility to confine light to the nanoscale and the ability to tune the dispersion relation of light have evoked large interest and led to rapid growth of plasmonic research. The parallel development of nanoscale fabrication techniques like electron beam lithography and focused-ion-beam milling has opened up new ways to structure metals' surfaces and control surface plasmon polariton propagation and dispersion at the nanoscale.

In 2000, Mark L. Brongersma *et al* (and others) proposed that EM en-

ergy could be transported below the diffraction limit with high efficiency and group velocity greater than $0.1c$ along a wire of its characteristic length 0.1λ . In year 2002, Maier *et al* experimentally observed the most efficient frequency for transport to be 3.19×10^{15} rad/sec with a corresponding group velocity of 4.0×10^6 m/s for longitudinal mode of plasmon waveguide having an inter-particle distance

of 75 nm. The achieved bandwidth was calculated to be 1.4×10^{14} rad/sec.

Dionne *et al* in year 2006 constructed slot waveguides. Slot waveguides can support both transverse electric and transverse magnetic photonic polarisation. The loss in slot waveguide can be minimised by using a low-refractive-index material; for example, a 100nm thick Ag/SiO₂/Ag slab waveguide sustains signal propagation up to 35 μ m at wavelength of 840 nm. In 2007, Feng *et al* observed that field localisation could be improved by introducing the partial dielectric filling of the metal slot waveguide, which also reduces propagation losses. The channel in metal surface waveguides supports surface plasmons at telecommunication wavelength with very low loss (having propagation length of 100 μ m) and well-confined guiding. In this experiment, surface plasmons are guided along a 0.6 μ m wide and 1 μ m deep triangular groove in gold material.